AdS GNN: PascalVOC

- We also tested it on a **segmentation task, PascalVOC-SP** from Long Range Graph Benchmark (Dwivedi et. al 2022)

- Given a point cloud of pixels as in right-hand panel, has to predict “what each pixel is” out of 20 classes (e.g. “sailboat”, “background”, etc.)

- Genuinely hard. Our model is again comparable to existing ones (e.g. the idea of the “shape” of an object is conformally invariant?)

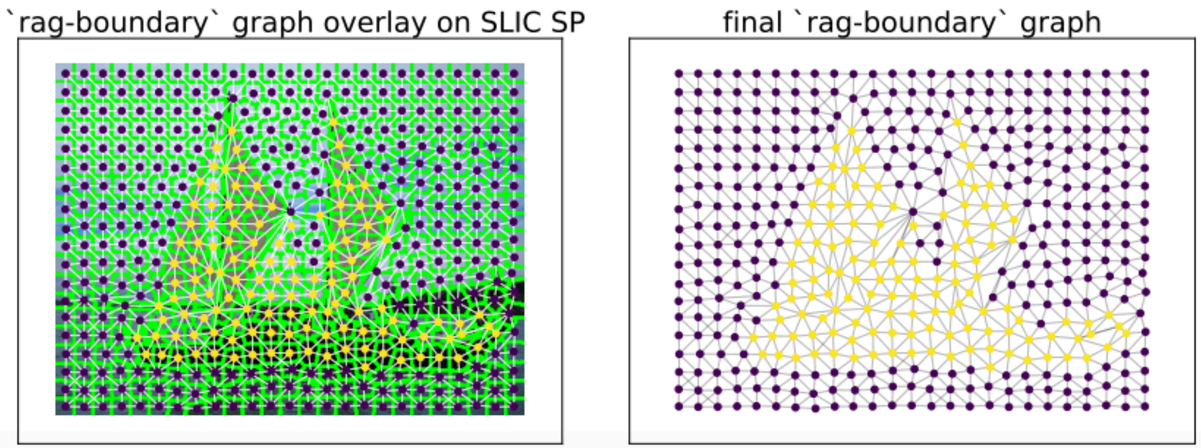
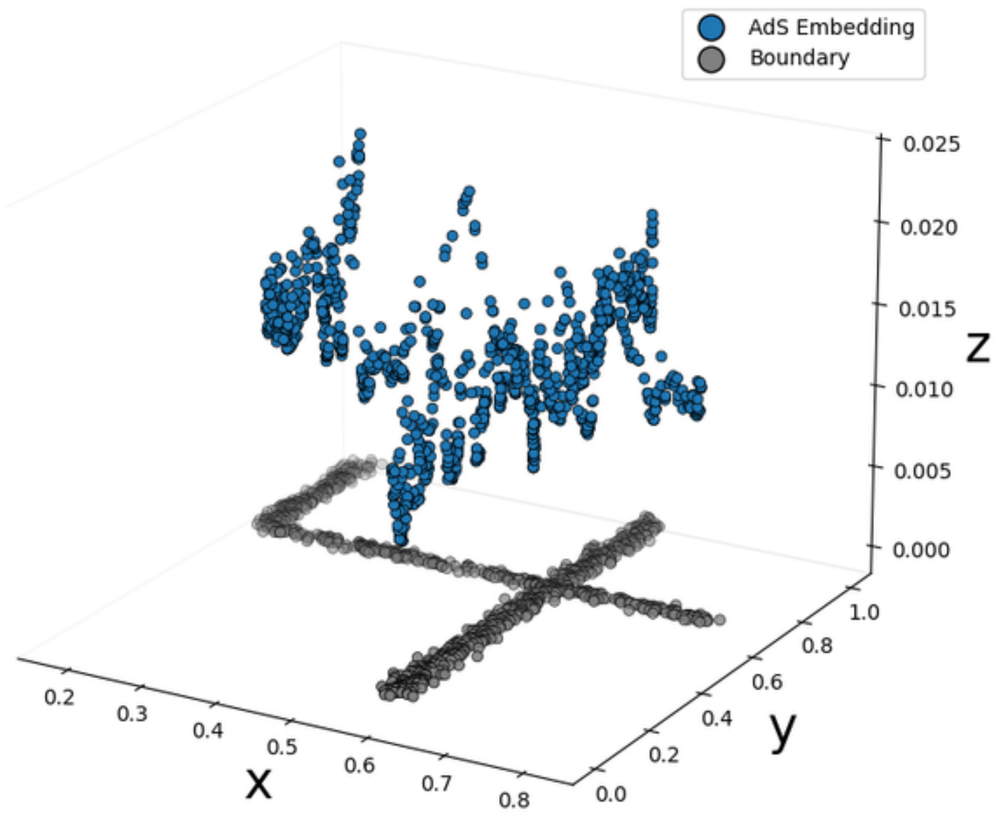


Table 2: Classification error on PascalVOC-SP.

Model	EGNN	AdS-GNN
Test F1 ↑	27.80 ± 0.74	28.07 ± 0.57

AdS-GNN Step 1: Details on Lifting

$$ds^2 = \frac{1}{z^2}(dz^2 + dx^2 + dy^2)$$



To embed a point into AdS, we need to pick a cutoff $z_0 \neq 0$.

1. Embed each point with a small finite $z_i = z_0$.
2. Find the k nearest neighbours of each point.
3. Compute their AdS "center of mass" (Galperin 1993) to get a $(\bar{x}_i, \bar{z}_i)$.
4. Embed the point as $X_i = (x_i, \bar{z}_i)$.

Physically: this sets the "length scale" of each point to be the average distance to its neighbours. (Does not depend much on initial z_0).